Global structure and the scales of exact realization

24 September 2026. Three continuations strengthen different parts of the first-line allowed-geometry/realization mechanism. They extend the preceding results, rather than repeat the integral calibration or replace the earlier obstruction theory. The four proof manuscripts below contain the arguments; this entry states their changes in scope and their relationship to Yau26.

1. Fiber thickness becomes a conclusion from global irreducibility

The thin-monodromy theorem removes the thick reference fiber from I32. Small actual conformal deformation around based generators implies a global alternative:

$$\boxed{\text{finite monodromy} \quad \text{or} \quad \text{one common invariant essential multicurve.}}$$

The constant depends only on fiber genus, not on generator count. The key point is simultaneous preservation: a gap among the finitely many short hyperbolic curve lengths supplies a single multicurve preserved by every generator. It is not an inference from individually reducible generators to reducibility of the group.

For a minimal Riemannian surface submersion with base and fiber genera at least two, let

$$\delta = \| \operatorname{Ric}^{\sharp} + I \|_{\infty,\mathrm{op}}, \qquad \eta = \|(K_B + 1)_+\|_{\infty}.$$

The complete implication is

$$2 \operatorname{diam}(B) \sqrt{\delta + \eta} < \mu_f, \qquad \text{irreducible monodromy}$$
$$\Longrightarrow \quad \boxed{M \text{ admits an exact negative KE metric.}}$$

Its universal cover is the hyperbolic bidisk. The output is on the original smooth bundle, by Nielsen realization and surface-bundle classification. The input does not assume a thick fiber, hyperbolic intrinsic fiber metrics, a flat connection, or actual gauge classes. It does assume the displayed global fibration and minimal submersion.

In the reducible case the multicurve has a genuine smooth realization through the entire bundle. A bounded-degree base cover labels its curves and sides and cuts the total space into surface bundles with boundary. Component representations and annular twists remain in the problem; they have not been shown to admit compatible Einstein matching.

The actual thin Dehn-twist family realizes this alternative, whereas signature-zero bundles with infinite irreducible monodromy obey a new thickness-independent base-scale lower bound. These are tests of the stated input theorem, not unrestricted Einstein nonexistence assertions.

2. Full nonlinear slow equations connect directly to actual gauge energy

The nonlinear slow-moduli theorem uses genuine horizontal hyperbolic fiber paths h_v, rather than linear tensors with an artificial quadratic amplitude error. Normalize the actual path tangent k_v by $\int_F |k_v|^2 = 1$, and put $G = \gamma + h_{v(x)}$. All Ricci blocks, including the normal acceleration of the path, are computed. The slow equation is exactly

$$\int_F \langle (\operatorname{Ric}_G + G)_{VV}, k_v \rangle = \frac{1}{2} \Delta_\gamma v.$$

For hyperbolic base its **full**, not projected, energy is

$$\boxed{\int_M |\operatorname{Ric}_G + G|^2 = \tfrac{1}{4}\int_B (\Delta v)^2 + \int_B \Xi(v)|dv|^4, \qquad \Xi(v) \geqslant \frac{1}{16\operatorname{Area}(F)}.}$$

The manuscript also gives the complete square decomposition for variable base curvature. Curvature adjustment cannot cancel the rank-one trace-free stress. Every exact Einstein metric in this entire block family has constant v and hyperbolic base.

The actual-gauge companion shows that the two canonical classes are actual monopole inputs and, for hyperbolic base, their harmonic forms are exactly $-(dA_B \pm dA_F)/(2\pi)$ at the representative level. There is no period error or zero set. Writing $r = \fint |\operatorname{Ric} + G|^2$ and $q_\pm = \mathfrak{D}_{2,\pm}/V$, it proves

$$\boxed{q_+ = q_- = \frac{9}{4V}\int_B |dv|^2 + \frac{9}{2}r = \frac{9}{2}\left(\fint |\nabla \omega_{\pm}|^2 + r\right).}$$

Thus the first-line functional defect measures an actual global moduli energy in this family. With variable base curvature the exact formula retains both the base-curvature variance and the nonzero negative projection of the canonical classes. In this family pointwise negative Ricci pinching itself implies $q_\pm \leqslant 54\delta + 9\delta^2$; no auxiliary-form existence assumption remains.

On genuine cyclic covers, fixed $a > 0$ and $v_m = a\cos(2\pi z/m)$ give

$$\delta_-(G_m) = \Theta(m^{-2}), \quad \mathscr{E}(G_m) = \Theta(m^{-4}), \quad q_{\pm,m} = \Theta(m^{-2}), \quad \int |\operatorname{Ric} + G_m|^2 = \Theta(m^{-3}),$$

while $\operatorname{Var}(v_m) = a^2/2$ stays fixed. The metrics remain a fixed distance from every fiber-preserving product output. The scalar scale $\lambda_1(B_m) \asymp m^{-2}$, not a local curvature spike or missing gauge field, accounts for this behavior.

This extends the previous shrinking-amplitude test but has a different scope: it does **not** prove fixed-distance separation from all Einstein metrics and arbitrary diffeomorphisms. The smooth products already have Einstein outputs. The complete scalar equation and retained normal term, rather than that unproved stronger exclusion, are the new nonlinear tool.

3. Original defects now give a quantitative test for whole-core zeros

The harmonic-zero scale theorem replaces I34's qualitative nonvanishing premise by a quantitative metric-scale test. For the actual self-dual harmonic class representative H, inherited form supply gives

$$\int_M \left(|-2\pi H| - \sqrt{2}\right)^2 \leqslant 818(r+q)V.$$

On a coordinate ball of radius ρ with fixed rescaled $C^{1,\alpha}$ metric control, the first-order Hodge equation gives

$$H(x) = 0 \quad \Longrightarrow \quad \int_{U_{x,\rho}} \left(|-2\pi H| - \sqrt{2}\right)^2 \geqslant c\rho^4.$$

No derivative of Ricci or extra pointwise bound on H is assumed. If every point of a **whole** singular filling has such a supplied coordinate scale at least ρ_{core}, then

$$(r+q)V < c_0\rho_{\mathrm{core}}^4 \quad \Longrightarrow \quad H \text{ has no zeros on the core.}$$

Under the retained A1/Betti or rational-ball hypotheses this forces the whole filling to be the appropriate **boundary-marked** Milnor fiber; the rational-ball case forces Wahl type. A separate global Q-Gorenstein smoothing then yields negative KE on the same smooth manifold. The coordinate scale may tend to zero; the comparison uses total defect, not merely its average.

A classical zero-circle birth, rescaled on a fixed negative KE product, provides actual harmonic representatives of the same canonical class with unchanged periods and arbitrarily small average form errors but persisting zeros. It proves that those form data alone are insufficient. Its pointwise Ricci defect is large, so it is not a counterexample to automatic matching under the full near-Einstein equation. Whether that equation itself forces the displayed core-scale ratio remains open.

4. Mathematical position and retained work

The first result enlarges an exact existence input class. The second solves a genuine nonlinear slow family and identifies its gauge energy and sharp global scale. The third gives a raw-defect-to-marked-filling criterion, with a model explaining why a scale is needed. These are three result families, not one universal obstruction/existence theorem.

Earlier uniform integral obstacles, parameterized non-saturation, two-chirality topology, and their actual gauge suppliers remain intact. Neither these continuations nor the absence of an obstruction settles general Einstein existence on the shear manifolds. They do not finish a complete Yau26 sign branch. The next unresolved information is now explicit: residual component/twist matching, non-adapted slow-mode coverage, and the core defect/scale relation under the full Ricci equation.

Reproducibility: bounded checks and proof-integration record. Local source commits and the exact checked scope are recorded there.